

Akash Rane , MS

New York City, NY | akashrane2609@gmail.com | +1(201)9188710 | [LinkedIn](#) | [GitHub](#) | [Kaggle](#) | [Portfolio](#)

SUMMARY

Data Scientist and ML engineer with an M.S. in Computer Science and 3+ years of experience building AI/ML systems, RAG pipelines, and production-oriented software applications. At Shoptaki, developed ML-based KYC verification, LangChain-powered RAG workflows, and scalable microservices for identity and risk decision systems. Previously built real-time Python/IoT analytics solutions at Optify, including sensor-data pipelines, monitoring dashboards, and on-site system deployments. Comfortable and proficient with a wide range of AI tools, coding assistants, and AI agents, using them extensively across development, research, analysis, and automation.

SKILLS

Data Science & ML: Python, Pandas, NumPy, Scikit-learn, Feature Engineering, Hyperparameter Tuning, Model Evaluation.

Generative AI & RAG: LLMs, RAG, LangGraph, Prompt Engineering, Embeddings, BM25, Hybrid Retrieval, Cross-Encoder Reranking.

Data & Analytics: SQL, PostgreSQL, Power BI, Matplotlib, Exploratory Data Analysis, Data Cleaning & Preprocessing.

AI Engineering & Cloud: FastAPI, REST APIs, Supabase, Docker, AWS, Apache Kafka, Git.

PROFESSIONAL EXPERIENCE

Shoptaki

New York

Data Scientist

Jan2026 -Present

- Designed an end-to-end AI-based KYC verification pipeline using Python and Scikit-learn, integrating structured and unstructured identity data with ML risk scoring to automatically classify users into auto-verify, human-review, or auto-reject workflows.
- Built a Retrieval-Augmented Generation (RAG) pipeline using LangChain to extract information from identity documents, retrieve verification context, and cross-reference user records against KYC rules for automated decision support.
- Engineered ML classification models for high-risk profile detection and integrated model outputs into downstream application workflows for automated decision support.
- Architected a 5-service microservices backend (gateway, auth, user, messaging, master) supporting WebSocket chat, WebRTC audio/video calling, and AES-256-GCM encrypted storage.

Optify Industrial Solutions Pvt. Ltd.

Pune, India

Data Engineer and Analyst

Jul 2022 – Dec 2023

- Led end-to-end design and development of a Python-based IoT analytics platform liaising with factory owners to gather requirements, implementing OPC-UA and MQTT protocols for real-time sensor data transmission, and delivering Node-RED and Grafana dashboards that significantly improved operational oversight.
- Managed a 1-month on-site deployment as primary technical and client-facing lead, configuring interfaces, debugging systems, and ensuring 100% operational integration and project sign-off.
- Created data-backed presentations on operational efficiency gains and presented findings to senior government stakeholders, securing innovation grant approvals demonstrating the ability to translate complex technical insights for non-technical executives.

PROJECTS *(All projects are available on my GitHub click the project names to explore the repositories.)*

F1 Pit-Stop & Driver Position Prediction | Scikit-learn, FastF1/OpenF1, GitHub Actions

- Built end-to-end ML pipelines using historical race, tyre, weather, and driver data to predict finishing positions and pit-stop strategies with leakage-safe feature engineering and chronological validation.
- Compared and evaluated machine-learning models using future-season holdouts and performance metrics, while building automated data pipelines spanning verified F1 datasets with 7K+ views and 1K+ downloads.

Failure—Pattern Retrieval Assistant | Python, Chroma, OpenAI, BM25, RAG

- Built a hybrid RAG system combining semantic embeddings, BM25, Reciprocal Rank Fusion, and cross-encoder reranking to retrieve similar equipment failures and generate citation-backed root-cause recommendations.
- Developed failure-mode evaluation using Precision@K and confidence-based pattern analysis to measure retrieval quality and identify recurring systemic failure patterns.

Job Hawk — AI-Powered Job Application Agent | Python, FastAPI, LangGraph, Llama, Gemini, Supabase

- Built an AI agent that parses resumes, discovers jobs, and scores candidate-job fit across skills, experience, domain relevance, and role alignment using LLM-driven structured evaluation.
- Orchestrated resume analysis, job ranking, and tailored application generation through LangGraph workflows with FastAPI and Supabase-backed human approval and application tracking.

NLP-Powered Power BI Custom Visual | TypeScript, D3.js, Gemini 2.5 Flash

- Built an NLP-powered Power BI visual that translates natural-language analytics questions into dynamically generated charts using intent classification, fuzzy field matching, and Gemini fallback.
- Developed validation and field-matching logic that maps user queries to appropriate dataset dimensions and measures while preventing invalid or misleading chart generation.

FinStream — Real-Time Financial Analytics Platform | Python, PostgreSQL, WebSockets, REST APIs, Docker, AWS

- Built real-time financial data pipelines to ingest, normalize, store, and serve multi-source market data for portfolio tracking, market analysis, and downstream analytical workflows.
- Developed transformations for price movement, portfolio performance, gain/loss, and market metrics while supporting low-latency delivery through APIs, persistent storage, and streaming architecture.

EDUCATION

Pace University, Seidenberg School of Computer Science and Information Systems

New York, NY

Master's in Computer Science | GPA: 3.89/4.00

Dec 2025

SPPU - Savitribai Phule Pune University

Pune, India

Bachelor of Engineering in Computer Engineering | **Concentration:** Data Science | **GPA: 8.18/10**

Jul 2022